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(1) Research Program/Project/Study Title: SAGIP INDIGENOUS PEOPLE COMMUNITY THROUGH DOST CEST IN DUMARAO, CAPIZ	
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Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component Title:	Component 1. Livelihood Enhancement to IPs Community
Component Leader:	Dr. Maryneth B. Barrios
Academic Rank:	Associate Professor V
Component Title:	Component 2: Environmental Protection and Conservation
Component Leader:	Mrs. Ness Marie Secretario
Academic Rank:	Associate Professor I
Component Title:	Component 3: Disaster Risk Reduction and Management and Climate Change Adaptation
Component Leader:	Mrs. April Joy Abelito
Academic Rank:	Assistant Professor 4
Component Title:	Component 4: Health and Nutrition
Component Leader:	Dr. Marie Jean Penson
Academic Rank:	Associate Professor V
Component Title:	Component 5: Enhancement of the IPs Education
Component Leader:	Jane P. Tropico
Academic Rank:	Assistant Professor 4

(2) Category (please check) ☐ Social Science/Education/Humanities & other related fields ☐ Natural/Biological Sciences	(3) Research Center [Check the concerned center] ☐ SSITRDC
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☐ Food Research/ ☐ Developmental/Computer/Engineering/Architecture ☒ Extension		Social Science and IndustrialTechnology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		Dumarao
Province:		Capiz
Region:		6
(5) Classification		(6) Mode of Implementation
Research: ☒ Basic ☐ Applied	Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	☐ Single Authorship/Proponent ☒ Collaborative Research(within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity		(8) Discipline/Field of Specialization
Livestock , IP Community		

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(9) Rationale

The Philippines is a multi-cultural country with over 169 living ethnolinguistic groups and about 140 of which are acknowledged as indigenous (Hirtz, 2003). These indigenous peoples represent 10 to 20% of the total Philippine population. Collectively, indigenous populations in the Philippines have been estimated at 12 million and are broadly classified as Lumad, Ilongot, Palawan, Mangyan, and Negrito (Respicio et al., 2008)

Categorized as Negrito, Ati people are the first local settlers of the Panay Island which consists of four provinces: Aklan and Capiz in the North, Antique in the West and Iloilo in the Southeast. Depending on what part of the Philippines, Negrito people are called using different names such as Agta, Ita, Ata, Aeta, Batak and Ati. One theory says that the Ati belong to the Old Stone Age Negrito stock, related to African pygmies, which migrated from the neighboring Asian countries by crossing land bridges exposed during the last glacial period about 25,000 years ago (Baruah, 2000).

In the province of Capiz, specifically in the municipality of Dumarao, this indigenous group resides in the mountainous areas of Barangay Aglalana and Tamulalod. They are still living using their traditional ways. Their production of brooms is the primary source of their income. Along with planting root crops and raising livestock, the other sources of income of Ati people include farming, hunting and charcoal making (Hirai, 2015).

Aside from livelihood, the Ati communities in Dumarao are facing several problems which hinder them from progressing. These include lack of education, threats on maternal and child's health, inadequate sources of food and water, bullying, cultural preservation in terms of their way of living, and being unprepared during the times of natural disasters. Hence, since Capiz State University – Dumarao Satellite College is an institution that is located near these two Ati communities, CAPSU – Dumarao will conduct extension services in order to help in addressing their problems through different interventions namely Livelihood/Economic Enterprise Development, Health and Nutrition, Environmental Protection and Conservation, Human Resource Development, and Disaster Risk Reduction and Climate Change Adaptation.

(9-A) Project Description

Issues/ Problems/ Identified Objectives

Impact Paradigm	Problem	Objectives
Socio Economic Impact	No sustainable sources of income by the IP community	To provide sustainable livelihood for the indigenous people community.
Environmental Impact	Deforestation and landslide threat through kaingin system. (Conversion of the forest into corn plantation).	To help the community for the environmental protection and conservation.

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	Threats of natural disaster to the community.	To increase preparedness and awareness on Disaster Risk Reduction Management (DRRM) and disseminate IEC materials
Health and Nutrition	Malnutrition and lack of nutritious foods for the community	To help the community to have nutritious foods available.
Education	Low academic assessment result of some of the IPs children. Lack of awareness of the financial literacy.	To conduct literacy and numeracy for IPs children. To conduct financial literacy to every heads of the family in the community.

Proposed Solutions/ Interventions

Objectives	Proposed Intervention/ Solution
To provide sustainable livelihood for the indigenous people community.	Training on Darag Native Chicken Production Acquisition and provision of the basic needs of Darag native chickens. Dispersal of Darag native chicken.
	Training on crop production using organic production.

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	Acquisition and provision of the basic needs of crop production.
	Construction of the housing for the high value crops.
To help the community to have nutritious foods available.	Conduct one (1) training on Pinggang Pinoy and Meal Management Program to at least (20) women of the IPs community.
To help the community for the environmental protection and conservation.	Conduct of one (1) training/seminar on the agroforestry and crop production technology. Conduct tree planting activity to the community.
To increase preparedness and awareness on Disaster Risk Reduction Management (DRRM) and disseminate IEC materials	Training/workshop on disaster preparedness and awareness. Dissemination of IEC materials
To conduct literacy and numeracy for IPs children.	Avail starbooks from DOST for Aglalana Integrated School.
	Conduct literacy and numeracy for IPs children.
To conduct financial literacy to every heads of the family in the community.	Conduct financial literacy to every heads of the family in the community.

(11) Review of Literature/Prior Art Search

The Philippines is a culturally diverse country with an estimated 17 million Indigenous Peoples (IPs) belonging to 110 ethno-linguistic groups in 2010. They are mainly concentrated in Luzon (33%), especially in the Cordillera Administrative Region; and Mindanao (61%) with some groups in Visayas area (IWGIA, 2011).

These indigenous peoples represent 10 to 20% of the total Philippine population. Collectively, indigenous populations in the Philippines have been estimated at 12 million and are broadly classified as Lumad, Ilongot, Palawan, Mangyan, and Negrito (Respicio et al., 2008)

Categorized as Negrito, Ati people are the first local settlers of the Panay Island which consists of four provinces: Aklan and Capiz in the North, Antique in the West and Iloilo in the Southeast. Depending on what part of the Philippines, Negrito people are called using different names such as Agta, Ita, Ata, Aeta, Batak and Ati (Baruah, 2000).

Capiz had a total of 720,000 population in 2010 with an IP population of 25,200. The main income sources are broom making, farming, livestock breeding, hunting, labor in non-IP farms,

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charcoal making, handicraft (buon buon) making, orchid and herbal medicine gathering. Although they have many sources of livelihood, the estimated average income of Ati women is from 600 to 2,000 pesos per month, which is merely about 6 to 21 percent of the average household income of Region 6 (Hirai, 2015).

Health issues like food security, water, and sanitation are also prevailing within Ati communities. Many eat only once or twice a day. The meal consists of rice and vegetables. As a consequence, nutritional problems exist. Inadequate sources of water and the distance of the available sources are the biggest problems for Atis. Aside from the distance of available water sources, the quantity of the water to be transported is also limited. This affects the hygienic practices. They wash their clothes and take a bath only 1-3 times a week. None has a sanitary toilet. Many have an open-pit toilet, and the rest do not have any toilet facilities at all (Hirai, 2015).

Aside from livelihood and health, one of the main problems of Atis is education. The Ati children hike a hilly and rugged trail to reach the school, which is one of the reasons why half of the school age population is not studying. Few of children aged 3-5 years old are enrolled in a Day Care Center. The major reason for dropping out or not attending school is financial. The Ati parents cannot afford school supplies, such as pencils and notebooks (Hirai, 2015).

On the other hand, deforestation has also caused environmental damages such as landslides and soil erosion. This ecological problem affects many people residing in the mountain areas, many of which are the Ati. Considering the Ati's dependence on the mountains for resources, deforestation has greatly affected the Ati's way of life. Farming and agriculture are now a big source for their sustenance. However, since most of the farmlands are owned by the Neo-Malays (the lowlanders coming from the lineage of Malay), most Ati resort to cultivating the flat tops of the mountains or farms in small river valleys. In their farming system, they practice a "slash and burn" system called caingin in which by-products of palay or new growth of *Cynodon Dactylon* grass or cogon are burned to make space for rice planting. This practice is said to be harmful for the environment (Maohong, 2012).

Since most of the Ati communities in Capiz, specifically in Dumarao, are located in mountainous areas, they are prone to various risks during natural calamities like typhoons, earthquakes, landslides, etc. These natural calamities can have significant effects to their lives. This effects may include the following: disruption of their livelihood such as agriculture, hunting, broom making and livestock raising, which can lead to food insecurity; being forced to relocate due to the destruction of their homes and other infrastructure, leading to displacement and social dislocation; lack of access to humanitarian aid, emergency services, and government services due to geographical isolation and cultural differences; loss on biodiversity and ecosystem services which they rely on for sustenance and traditional medicines; and exposure to several diseases, malnutrition, and limited access to healthcare services.

Department of Science and Technology (DOST) is the premier science and technology governing agency in the country with the mandate of providing central direction, leadership, and coordination of scientific and technological activities; as well as formulating policies, programs, and projects to support national development (DOST, 2017).

CEST or Community Empowerment through Science and Technology is a program of DOST that is targeting to empower the poorest and most depressed communities in the country, via S&T interventions in health and nutrition, water and sanitation, basic education and literacy,

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livelihood/economic enterprise development, and disaster risk reduction and climate change adaptation (Lopez, 2015).

Methodology

Capiz State University in partnership with DOST VI, NCIP VI, and IP communities' leaders will collaborate in the planning, project implementation, monitoring and evaluation of the activities to be conducted.

The general project methodology or activities is outlined below:

- **Reconnaissance and Focus Group Discussion.** With the initial discussion conducted by Capiz State University with Capiz PSTO through the LGU and the community leaders of Ati and Panay Bukidnon, needs and problems were identified, and possible solutions were discussed.
- **Evaluation of Gaps and Identification of Possible S&T Interventions.** These may be technical, socio-cultural, managerial, and marketing. It may also be related to the source of agricultural inputs or raw materials. S&T interventions should cater to the identified gaps and a part of answering the general and specific objectives. Social acceptability is also important, considering that S&T interventions should be acceptable to the targeted beneficiaries.
- **Capacity Building.** Activities will either be creative workshops, hands-on learning, or guided focus group discussions. Technical matters will involve demonstrations and hands on skill-building. Ultimately, the initial beneficiaries will be expected to provide capacity building and awareness to other individuals upon their reach.
- **Economic Analysis.** This can translate the outcome of the S&T interventions into economic benefits that can convince other interested individuals.
- **Policy Initiation, Development, and Advocacy.** This will help the community make itself self-sufficient and sustainable. Policies will be implemented to make the community well-developed and compass its advocacy.
- **Sustainability Planning.** This will provide Dumarao IP communities a road map for achieving long-term goals and document strategies to continue the program, activities, and partnerships.

Component 1. Livelihood Enhancement to IPs Community

Two livelihood programs will be introduced to the community: Darag native chicken production and High value crop production. The output of these projects will provide the additional income for the beneficiaries.

Component 2: Environmental Protection and Conservation

This component will give emphasis on the deforestation problem for the community through seminar for agroforestry and crop integration, tree planting activity, waste management and vermicomposting.

Component 3: Disaster Risk Reduction and Management and Climate Change Adaptation

Improved disaster awareness will help the community prepare for the upcoming disaster and equipped with the knowledge of what to do before and during a disaster.

Component 4: Health and Nutrition

A nutrition tool as a guide in daily meal planning, the Pinggang Pinoy developed by DOST-FNRI will be shared with mothers in selected mother beneficiaries through training. The information and knowledge acquired by the mothers will result in improved health and nutrition of families and prevent malnutrition in children. Furthermore, the training will provide information on how to prepare nutritious foods for households at low-cost expenses.

Component 5: Enhancement of the IPs Education

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To help IPs students to improve their education, Starbooks will be put in the nearby school where most of the IPs students are spent their school days. Literacy and numeracy training will also done to all children to improved their academic performance.

(14) Major Activities/Work plan

Objectives	Target Activities	YEAR 1			
		Y1	Y2	Y3	Y4
To provide sustainable livelihood for the indigenous people community.	Training on Darag Native Chicken Production				
	Acquisition and provision of the basic needs of Darag native chickens.				
	Dispersal of Darag native chicken.				
	Training on crop production using organic production. Acquisition and provision of the basic needs of crop production. Construction of the housing for the high value crops.				
To help the community to have nutritious foods available.	Conduct one (1) training on Pinggang Pinoy and Meal Management Program to at least (20) women of the IPs community.				

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To help the community for the environmental protection and conservation.	Conduct of one (1) training/seminar on the agroforestry and crop production technology. Conduct tree planting activity to the community.				
To increase preparedness and awareness on Disaster Risk Reduction Management (DRRM) and disseminate IEC materials	Training/workshop on disaster preparedness and awareness. Dissemination of IEC materials				
To conduct literacy and numeracy for IPs children.	Avail starbooks from DOST for Aglalana Integrated School.				
	Conduct literacy and numeracy for IPs children.				
To conduct financial literacy to every heads of the family in the community.	Conduct financial literacy to every heads of the family in the community.				

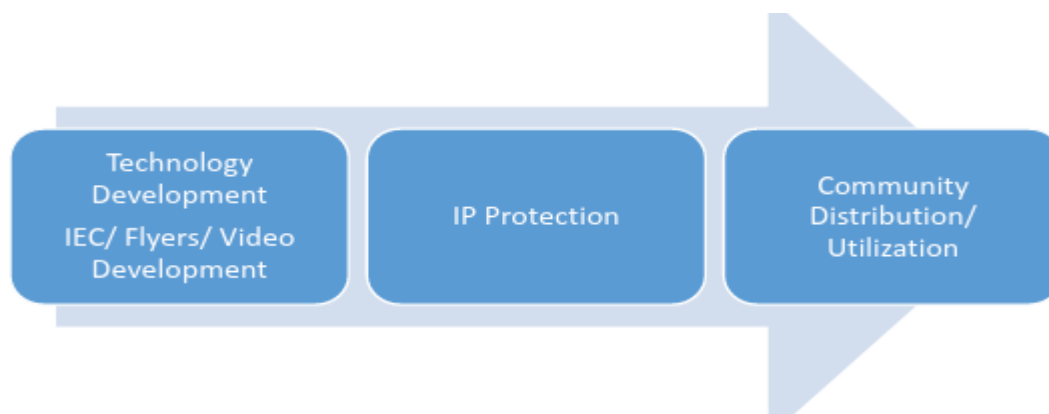
(15) Expected Output

EXPECTED OUTPUTS	DETAILS	TARGET
People Services	Training of IP communities for livelihood project Enhance IP children education	50
Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials	2
Patents/IP	File 2 copyrights for the developed IEC materials and videos.	2

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Places and Partnerships	Forge linkages and partnerships to LGU's, NCIP, Capiz Tourism Office, Dumarao LGU, and other organizations.	5
Products	Sustainable DNC and high value crop production	
Policy	Develop one policy inputs for the sustainable livelihood project for the IP communities	1
Impact to Social/ Economics	<i>Social Impact:</i> <i>Economic Impact:</i> Increase IP communities income	

(15) Technology Transfer Activity Design



(16) Targets Beneficiaries

- Ati Community
- NCIP
- Department of Tourism

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Dr. Maryneth B. Barrios	Study Leader	Animal Science
Dr. Marie Jean P. Penson	Member	
Ness Marie Secretario		
April Joy Abelito		
Jane Tropico		
Nelinda Duero		

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount

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A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE)		
Transportation allowance		
Representation Expenses		
Training Expenses		
Supplies (Office supplies and Foods/ Give aways to Ati Respondents)		
Total		2,000,000.00

(19) Literature Cited

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